

Elevation of boiling point of the solvent

- The latent heat of vapourisation of water is 9700 Cal/mole and if the *b.p.* is 100°C , ebullioscopic constant of water is
 - 0.513°C
 - 1.026°C
 - 10.26°C
 - 1.832°C
- The molal elevation constant of water = 0.52°C . The boiling point of 1.0 molal aqueous *KCl* solution (assuming complete dissociation of *KCl*), therefore, should be
 - 100.52°C
 - 101.04°C
 - 99.48°C
 - 98.96°C
- The rise in the boiling point of a solution containing 1.8 gram of glucose in 100g of a solvent is 0.1°C . The molal elevation constant of the liquid is
 - 0.01 K/m
 - 0.1 K/m
 - 1 K/m
 - 10 K/m
- If 0.15g of a solute dissolved in 15g of solvent is boiled at a temperature higher by 0.216°C than that of the pure solvent. The molecular weight of the substance (molal elevation constant for the solvent is 2.16°C) is
 - 1.01
 - 10
 - 10.1
 - 100
- Pressure cooker reduces cooking time for food because
 - Heat is more evenly distributed in the cooking space
 - Boiling point of water involved in cooking is increased
 - The higher pressure inside the cooker crushes the food material
 - Cooking involves chemical changes helped by a rise in temperature
- Which of the following statements is correct for the boiling point of solvent containing a dissolved solid substance
 - Boiling point of the liquid is depressed
 - Boiling point of the liquid is elevated
 - There is no effect on the boiling point
 - The change depends upon the polarity of liquid
- When a substance is dissolved in a solvent, the vapour pressure of solvent decreases. It brings
 - A decrease in boiling point of solution
 - An increase in boiling point of the solution



- (c) A decrease in freezing point of the solution
(d) An increase in freezing point of the solution
8. Elevation in boiling point was 0.52°C when 6gm of a compound X was dissolved in 100gm of water. Molecular weight of X is (K_b for water is 0.52 per 1000 gm of water)
(a) 120 (b) 60
(c) 180 (d) 600
9. If the solution boils at a temperature T_1 and the solvent at a temperature T_2 the elevation of boiling point is given by
(a) $T_1 + T_2$ (b) $T_1 - T_2$
(c) $T_2 - T_1$ (d) $T_1 \div T_2$
10. If for a sucrose solution elevation in boiling point is 0.1°C then what will be the boiling point of NaCl solution for same molal concentration
(a) 0.1°C (b) 0.2°C
(c) 0.08°C (d) 0.01°C
11. The molal elevation constant is the ratio of the elevation in B.P. to
(a) Molarity
(b) Molality
(c) Mole fraction of solute
(d) Mole fraction of solvent
12. The molal boiling point constant for water is $0.513^{\circ}\text{C kg mol}^{-1}$. When 0.1 mole of sugar is dissolved in 200ml of water, the solution boils under a pressure of one atmosphere at
(a) 100.513°C
(b) 100.0513°C
(c) 100.256°C
(d) 101.025°C
13. Value of gas constant R is
(a) 0.082 litre atm
(b) $0.987\text{ cal mol}^{-1}\text{K}^{-1}$
(c) $8.3\text{ J mol}^{-1}\text{K}^{-1}$
(d) $83\text{ erg mol}^{-1}\text{K}^{-1}$
14. The temperature, at which the vapour pressure of a liquid becomes equal to the atmospheric pressure is known as
(a) Freezing point
(b) Boiling point
(c) Absolute temperature
(d) None of these
15. The elevation in boiling point of a solution of 13.44g of CuCl_2 in 1kg of



- water using the following information will be
(Molecular weight of $\text{CuCl}_2 = 134.4$ and $K_b = 0.52 \text{ K molal}^{-1}$)
- (a) 0.16 (b) 0.05
(c) 0.1 (d) 0.2
16. When 10g of a non-volatile solute is dissolved in 100 g of benzene, it raises boiling point by 1°C then molecular mass of the solute is (K_b for benzene $= 2.53 \text{ K molal}^{-1}$)
(a) 223 g (b) 233 g
(c) 243 g (d) 253 g
17. An aqueous solution containing 1g of urea boils at 100.25°C . The aqueous solution containing 3 g of glucose in the same volume will boil at (Molecular weight of urea and glucose are 60 and 180 respectively)
(a) 100.75°C (b) 100.5°C
(c) 100.25°C (d) 100°C
18. When common salt is dissolved in water
(a) Melting point of the solution increases
(b) Boiling point of the solution increases
(c) Boiling point of the solution decreases
(d) Both melting point and boiling point decreases
19. During the evaporation of liquid
(a) The temperature of the liquid will rise
(b) The temperature of the liquid will fall
(c) May rise or fall depending on the nature
(d) The temperature remains unaffected
20. At higher altitudes the boiling point of water lowers because
(a) Atmospheric pressure is low
(b) Temperature is low
(c) Atmospheric pressure is high
(d) None of these
21. The elevation in boiling point for one molal solution of a solute in a solvent is called
(a) Boiling point constant
(b) Molal elevation constant
(c) Cryoscopic constant
(d) None of these
22. A solution of 1 molal concentration of a solute will have maximum boiling point elevation when the solvent is
(a) Ethyl alcohol



- (b) Acetone
(c) Benzene
(d) Chloroform
23. Mark the correct relationship between the boiling points of very dilute solutions of $BaCl_2(t_1)$ and $KCl(t_2)$, having the same molarity
- (a) $t_1 = t_2$
(b) $t_1 > t_2$
(c) $t_2 > t_1$
(d) t_2 is approximately equal to t_1

